Gazi Tahsina Sharmin Jahin

Computer Science & Engineering Graduate

Email | **LinkedIn** | **GitHub** | **Portfolio** | **Google Scholar** | **Kaggle**

Research Focus:

- Explainable Machine Learning
- Deep Neural Networks
- Trustworthy & Robust AI Systems
- Federated & Privacy-Preserving AI
- Computer Vision
- AI in Healthcare

Objective

Researcher in explainable and trustworthy AI, focused on developing interpretable models that remain reliable under real-world challenges. My goal is to advance transparent and robust machine learning, building dependable computer vision systems for complex applications.

Research Experience

Certificate

- **Research Associate (Volunteer) - Artificial Intelligence Lab, IIUC** 05/2026–Present
  - Mentor undergraduate researchers in machine learning, NLP, and computer vision, guiding end-to-end research from literature review and experimental design to model development, evaluation, and manuscript preparation.
  - Currently developing an automated mango leaf disease diagnosis and severity estimation system using 40K augmented images across 8 disease classes, benchmarking 4+ pretrained CNN architectures.
- **Research Assistant (Full-Time) - GETE Research Center - Dhaka, Bangladesh** 01/2026–Present
  - Conducted coastal fieldwork near Kumira Beach, Chattogram, collecting algae, surface-soil samples, & continue to support interdisciplinary research through systematic data organization, analysis, & documentation.
- **Research Assistant (Part-Time) — Machine Learning & Data Mining Lab, IIUC** 04/2024–12/2024
  - First author of a peer-reviewed journal publication in the *International Journal of Innovative Computing.*
  - Curated a dataset of 5,000 Bengali fake news articles, implemented a hybrid classification pipeline achieving 97% mean accuracy. Applied LDA for topic modeling and integrated LIME to ensure transparency of model predictions.

Education

- **Bachelor of Science (BSc)**, *Computer Science and Engineering*
  International Islamic University Chittagong - CGPA: 3.602 out of 4.00
  Ranked among the top 15% throughout all academic years 10/2019–12/2024

Relevant Coursework: Machine Learning & Data Mining, Neural Network & Fuzzy System, Linear Algebra, Software Engineering, Artificial Intelligence, Data Structures, Computer Algorithms, Differential & Integral Calculus, Probability & Statistics, Computer Security.

Undergraduate Thesis: ***(A+ Grade · Highest Marks in Thesis Defense)***
Bengali CyberBullying Detection using Sentiment Analysis and Topic Modeling: Insights with Advanced XAI Methods on NLP Models

- Developed a cyberbullying detection system for Bengali social media using BERT, ML, DL, and transformer models.
- Built and processed a dataset of 75,000+ Bengali texts with sentiment analysis and LDA topic modeling.
- Achieved up to 93% accuracy, with LIME-based XAI to interpret model predictions for moderation insights.

Tools: Google Colab, Python, Pandas, NLTK, Hugging Face Transformers, Matplotlib, Scikit-learn, TensorFlow. Code

Research Publications

Google Scholar

Journal Articles

*(*C = Conceptualization, M = Methodology, W = Writing(original draft), S = Software, D = Data Curation, F = Formal Analysis, I = Investigation, V = Validation, Vz = Visualization)*

1. *"Bengali cyberbullying detection: A comprehensive dataset for advanced analysis"*, Data in Brief, Elsevier, 2025 [First Author; Role: C,M,W,S] DOI
2. *"Enhancing fake news analysis in Bengali: A hybrid model with LDA, and LIME for interpretable classification"* International Journal of Innovative Computing, Universiti Teknologi Malaysia, 2025. [First Author; Role: C,S,M,W] DOI
3. *"Building Reliable Pediatric Voice Disorder Detection with Leakage Safe and Calibration Aware Machine Learning"*, International Journal on Advanced Science, Engineering and Information Technology (IJASEIT), 2026. (Under Review) [Second Author; Role: S,W,I]
4. *"AttnFusionNet: Learning Brain Patterns for Speech Delay Detection from rs-fMRI"*, Diagnostics, MDPI, 2026. (Submitted) [Second Author; Role: C,I,Vz,F]

Selected Conference Proceedings

First Author Publications

1. "Interpretable plant leaf recognition: Calibration and domain shift analysis with occlusion sensitivity and Grad-CAM," *in Proceedings of the 2025 11th IEEE International Women in Engineering (WIE) Conference on Electrical and Computer Engineering (IEEE WIECON-ECE)*, 2025. [Role: C,M,W,S] DOI
2. "Privacy-Preserving and Explainable Multi-Hospital CKD Prediction via Federated Learning with Blockchain Logging and Differential Privacy" *in Proceedings of the IEEE 2nd International Conference on Computing, Applications and Systems (COMPAS)*, 2025. [Role: C,M,S,W] DOI
3. "Flood resilient rice yield prediction in Bangladesh using explainable machine learning with prototype implementation," *in Proceedings of the 4th IEEE International Conference on Robotics, Automation, Artificial Intelligence and Internet-of-Things (RAAICON)*, 2025. [Role: C,M,W,S,D] DOI
4. "Quantifying and Predicting Violence Against Women Using the Violence Severity Index and Explainable Deep Learning," *in Proceedings of the 2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN), Scopus Indexed*, 2026. [Role: C,M,W,I] DOI

Co-Authored Publications

1. "EyeNet: An Explainable AI-Based Retinal Disease Classification by Using Hybrid Deep Learning Approach," *in Proceedings of the 2025 28th International Conference on Computer and Information Technology (ICCIT)*, 2025. [Role: M,Vz,F] DOI
2. "Federated Big Data Analytics for Student Performance Under Data Locality & Privacy Constraints," *in Proceedings of the International Conference on AI Innovations & Industry (ICAIII)*, 2026. [Role: C,D,W] DOI
3. "Robust and Explainable Facial Expression Recognition Using YOLOv10 Under Real-World Visual Degradations," *in Proceedings of the 2025 11th IEEE International Women in Engineering (WIE) Conference on Electrical and Computer Engineering (WIECON-ECE)*, 2025. [Role: M,S,W,Vz] DOI
4. "A Federated Learning Framework for Sustainable Boro Rice Yield Prediction Using Multi-Regional Agro-Climatic Data," *in Proceedings of the 2025 IEEE 2nd International Conference on Computing, Applications and Systems (COMPAS)*, 2025. [Role: D,C,I,W] DOI
5. "Explainable Comprehensive Analysis of Cross-dataset Phishing Detection Models," *in Proceedings of the 2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN), Scopus Indexed*, 2026. [Role: M,F,W] DOI
6. "Dengue sensitivity and vulnerability indices: An explainable machine learning approach to climate-health risks in Bangladesh," *in Proceedings of the 2025 International Conference on Intelligent Data Analysis and Applications (IDAA)*, Taylor & Francis, In Press, 2025. [Role: C,W,V]
7. "Unveiling Bengali text summarization: A powerful unsupervised approach with BanglaBERT, graph-based ranking, and interpretable insights," *in Proceedings of the 2025 IEEE International Conference on Quantum Photonics, Artificial Intelligence, Networking (QPAIN)*, Scopus Indexed, 2025. [Role: M,S,W,Vz] DOI
8. "An Explainable Hybrid Approach for Generation Z Slang Detection," *in Proceedings of the 2025 11th IEEE International Women in Engineering (WIE) Conference on Electrical and Computer Engineering (WIECON-ECE)*, 2025. [Role: F,V] DOI
9. "An Explainable AI-Driven Framework for Movie Recommendation System Using Big Data Analytics," *in Proceedings of the 11th International Conference on Information and Communication Technology for Intelligent Systems (ICTIS)*, In Press, 2026. [Role: D,V,F]

Academic Projects

GitHub

1. **BrainAI: Brain MRI Analysis & Knowledge Assistant** Code
   • Developed a medical AI research platform integrating a FastAPI backend, MRI analysis interface, and a source-aware RAG-inspired assistant for retrieving brain-tumor information using keyword-overlap ranking. • Implemented modular API endpoints for MRI upload handling and knowledge-base queries with source-attributed retrieval results. ***Tools:*** *Python, FastAPI, U-Net, HTML, CSS, JavaScript, REST APIs, Git, GitHub.*
2. **Dual-Stream Emotion Detection with YOLOv10** Code
   • Developed explainable facial expression recognition framework using dual-stream augmentation, YOLOv10, interpretability methods to evaluate model reliability under occlusion, blur, noise. • Achieved 0.93 mAP@50 with temperature scaling & calibration-based evaluation. ***Tools:*** *Python, YOLOv10, PyTorch, OpenCV, SHAP-CAM, RISE*
3. **Pneumonia X-Ray Analyzer** Code
   • Developed and deployed an EfficientNetB0-based chest X-ray classification application with a FastAPI inference backend and interactive Streamlit frontend, including configurable decision thresholds and experimental Grad-CAM explainability. ***Tools:*** *Python, TensorFlow, Keras, EfficientNetB0, FastAPI, Streamlit, NumPy, Pillow, Grad-CAM, Render.*
4. **Flood Resilience Dashboard** Code
   • Developed a Streamlit dashboard to calculate Flood Resilience Index (FRI), predict crop yields, and support division-level scenario analysis with global and local SHAP explanations. ***Tools:*** *Python, Scikit-learn, CatBoost, Ridge, RandomForest, SHAP, Pandas, NumPy, CSV export, Streamlit, Matplotlib.*

5. **Context-Aware Chatbot Suite with LangChain** Code
   • Developed a multi-page Streamlit chatbot suite using LangChain, supporting conversational memory, web-assisted queries, document-based question answering, SQL database interaction, and website content retrieval. ***Tools:*** *Python, LangChain, Streamlit, LLMs, SQL, Docker, REST APIs.*

Industrial Training

Certificate

Trainee–Robust Research And Development Ltd. 04/2024 – 09/2024
Project: Web Scraping and Summarization Platform using LLM

- Designed and developed a web-based platform to automate data collection from online sources using web scraping, with LLM-powered text summarization to transform raw content into concise, structured reports.
- Built interactive dashboards to present summarized insights, extracted information, supporting faster decision-making.
  Tools: *Scrapy, Tableplus, OpenAI, Streamlit, FastAPI, SQLAlchemy, Databases.* Code

Skills

- **Language Skills:** **Bangla** Native
  English **IELTS Academic: 8.00** Listening: 9.0, Reading: 8.0, Writing: 7.5, Speaking: 7.0
- **Digital Skills:** MS Office Suite — *Word, Excel, PowerPoint, Outlook, Access, Visio, Teams*
- **Technical Skills:**
  - **Programming:** *Python, NumPy, Pandas, Scikit-learn, Matplotlib, NLTK, MySQL, HTML, CSS*
  - **ML, Computer Vision:** *TensorFlow, Keras, PyTorch, OpenCV, YOLOv10, SHAP, LIME, Grad-CAM, LangChain*
  - **Research Tools:** *Streamlit, FastAPI, REST APIs, Scrapy, Docker, Google Colab, Jupyter Notebook, Git, GitHub, VS Code, PyCharm*
- **Content Creation & Writing:** *Canva, Adobe Photoshop, Adobe Illustrator, Google Docs, Grammarly*
- **Soft Skills:** Communication, Problem-Solving, Teamwork, Time Management

Awards and Achievements

Certificates

Awards & Honors

- **Research Excellence Grant**, International Islamic University Chittagong 2026
- **IELTS Excellence Scholarship**, IDP Education Bangladesh and The Daily Star 2025
- **Best Project**, Software Engineering Lab (CSE-3638), International Islamic University Chittagong 2023
- **Best Speaker – 1st Position**, Inter-departmental Parliamentary Debate, IIUC 2022
- **Semi-Finalist**, IIUC Competitive Programming Bootcamp, Spring 2021

Professional Activities & Development

- **Peer Reviewer**, Elsevier journal manuscripts 2026–Present
- **Adjudicator**, International Debate Competition, ELLS Society, Bangladesh 2024
- **Workshop on Generative AI and Large Language Models (LLMs)**, Bangladesh Computer Society 2026
- **Technical Workshops and Seminars in Machine Learning, Networking, and Emerging Technologies**
  IEEE Computer Society, IIUC Student Branch Chapter 2024–2025
- **Machine Learning and Data Science with Python Bootcamp**, Udemy 2021
- **Organizing Member**, IIUC Model United Nations 2021–2022
- **Volunteer**, Bangladesh Youth Advancement Institute 2021

Extracurricular Activities

View Activites

Vice-President, Graphic Design & Content Writing Team — IIUC Debaters' Community 2021–2024
• Led debate preparation and mentoring for junior members; participated in national & international competitions with distinction. • Organized university debate events, and adjudicated in academic debate platforms.

References

Dr. Muhammed Jamshed Alam Patwary
Lecturer, CSE Department
University of Chittagong
jamshed@cu.ac.bd

Mohammad Mahadi Hassan
Associate Professor, CSE Department
International Islamic University Chittagong
mahadi@iiuc.ac.bd

Dr. ABM Rafiqul Islam
Former Researcher, Department of Human Ecology
The University of Tokyo, Japan
geterafi2012@gmail.com